

SIH26038 — Gap Closure Verdict & Full Antigravity Build Prompt

RetinaScan AI / SightShield AI — Explainable DR Screening — Repo: THOUFIKUR/sih2026

1. Will the previous prompt satisfy the PS? — Honest Verdict

Implementing the previous prompt fixes the **integrity problems** in the repo — the fabricated benchmark script, the mislabeled Grad-CAM, and the missing training evidence. That is necessary and important. But it does not, by itself, fully satisfy every line of the PS. Some requirements depend on real-world outcomes (model performance, clinician judgment) that code cannot guarantee, and a few specific PS asks were not covered in the previous prompt at all. Both are addressed below.

PS Requirement	After previous prompt	Fully satisfied?
1. Image Quality & Enhancement	Already complete	YES
2. Retinal Structure Segmentation	Vessels + lesion candidates validated against real MESSIDOR-2 DRIVE code	NO — DRIVE code is still unvalidated, NV permanently heuristic
3. DR Severity Grading (>90%/85%)	Real evaluation script exists	DEPENDS ON MODEL — script reports truth, doesn't guarantee th
4. Explainability Module	Real Grad-CAM implemented	MOSTLY — calibration still data-limited, 30s claim unmeasured
5. Simulink Workflow	Tied to real pipeline timing	YES
"Outperforms any single technique"	Not addressed in previous prompt	NO — needs a real ablation study (added below)
"Sub-pixel" MA detection	Classifier stage added, but no sub-pixel refinement	NO — needs centroid refinement (added below)
Medical Imaging Toolbox usage	Not used anywhere in repo	NO — PS explicitly lists it as expected (added below)

2. Gaps No Prompt or Tool Can Close

- These are structural limits, not implementation gaps. Be upfront about them in your presentation rather than letting a judge discover them.
- **The performance threshold itself.** An honest evaluation script tells you the truth about your model — it cannot make the model better. If real sensitivity/specificity on Messidor-2 falls short of 90%/85%, the fix is retraining, better data augmentation, or ensembling — not a prompt.
 - **Neovascularization validation.** IDRI's lesion ground truth covers microaneurysms, hemorrhages, hard exudates, and soft exudates — not neovascularization. None of your four listed datasets (APTOS, IDRI, DRIVE, Messidor-2) contain NV masks. This module must stay labeled as an unvalidated research heuristic permanently, or you'd need an entirely different dataset not in the PS's list.
 - **The <30-second ophthalmologist validation claim.** This is a human-factors measurement. It requires literally timing a real (or at minimum, clinically-trained) reviewer looking at your report output. No code path produces this number — you'd need to run a small informal timing test yourself with a mentor, doctor contact, or even a structured self-timed walkthrough, and report it as pilot data, not claim it's clinically proven.
 - **Dataset access lag.** Messidor-2 requires signing ADCIS's data use agreement; IDRI requires an IEEE DataPort account. These aren't blocked by code — budget real time for approval before build day.

- **"Clinically meaningful" is a judgment call, not a metric.** Grad-CAM being real (gradient-based) is necessary but a judge or ophthalmologist ultimately decides if it's useful — you can't code your way to that adjective, only make the underlying method honest enough to deserve it.

3. What This Updated Prompt Adds (vs. the previous one)

- **Real ablation study** — trains/evaluates a bare single-technique classifier baseline alongside the full integrated pipeline on the identical test split, so "outperforms any single technique" becomes a real, reproducible comparison instead of a claim.
- **Sub-pixel MA centroid refinement** — adds quadratic/parabolic interpolation around each detected blob's intensity peak so centroid coordinates are reported at sub-pixel resolution, which is what the PS literally asks for and the previous version didn't implement.
- **OD/fovea ground-truth validation** — IDRiD's Localization subset includes real OD and fovea center coordinates. The previous prompt didn't use this. Now both modules get real pixel-distance-error validation instead of staying permanently unvalidated.
- **Medical Imaging Toolbox usage** — the PS explicitly lists this toolbox. Adds it for structured clinical image/report handling (e.g. DICOM-style structured metadata export for the annotated report), so the toolbox list in your submission isn't just decorative.
- **A pilot ophthalmologist-timing protocol** — not code, but a lightweight measurement plan you can actually run yourself before the demo, so the 30-second claim has at least honest pilot evidence behind it.

4. Full Updated Antigravity Prompt

Paste this as one block into Antigravity, copied to your repo root.

```
You are working inside my existing repo at [path to sih2026/retina]
I have verified the following by reading the actual source code myself —
treat these as confirmed facts, not assumptions:

- backend/routes/inference.py's generate_evidence_heatmap() (aliased as
generate_gradcam_heatmap/generate_scorecam_heatmap) does NOT compute
gradients or use feature maps. It is a CLAE + background-subtraction
+ YOLO-bbox-fusion heuristic. It has a dead, unused 'feature map' parameter.
- matlab/validation/evaluate_idr_benchmarks.m contains HARDCODED result
strings ("PASSED BENCHMARK (95.8%)") independent of its own computed
variables. Treat this file as fully invalid, not merely "synthetic".
- backend/training/train_dr_classifier.py is legitimate (real Cohen's
Kappa, real stratified split) but requires a parquet file at
data/samples/idrid/data/test-000000-of-000001.parquet that does not exist
in the repo. No training log anywhere proves what the deployed model
actually achieved.
- Only 5 IDRiD sample images and 1 DRIVE image/mask pair exist in data/.
No lesion ground-truth masks, no OD/fovea coordinates, no Messidor-2,
no APTOS bulk data are present yet.

Do NOT modify frontend/, backend/models/ (ONNX files), or Supabase config.

HONEST LIMITS — do not attempt to code around these, just implement the
measurement/reporting correctly and let the real result be whatever it is.
- Neovascularization has no ground truth in any of my four datasets
(APTOS/IDRiD/DRIVE/Messidor-2). detect_neovascularization.m must remain
permanently labeled as an unvalidated heuristic — do not add a fake
validation step for it.
- The PS's >90% sensitivity / >85% specificity target is an outcome of
real model quality, not something an evaluation script can force. Report
the real number, whatever it is, and do not adjust thresholds, retrain
selectively on test data, or cherry-pick a favorable split to hit the
target.
```

Your task, in this exact order:

1. DELETE evaluate idr benchmarks m entirely. Do not leave a version that could be mistaken for a real result.
2. Write matlab/data/loading/load idrid dataset m and load drive dataset m reading from data/idrid/ and data/drive/ (I will populate these myself from IEEE DataPort / DRIVE – do not fetch datasets yourself). These must fail loudly if folders are empty or missing expected files – never silently substitute synthetic data. load idrid dataset m must also load the localization subset (OD center coordinates, fovea center coordinates) as a separate return value.
3. Write backend/utils/true_gradcam.py implementing REAI gradient-based Grad-CAM: forward hook on the model's last convolutional layer backprop the gradient of the predicted class score w.r.t. that layer's activations, weighted activation map, unsample, overlay. load the original PyTorch checkpoint for this gradient pass (ask me for its path if not in the repo). Rename the existing generate_evidence_heatmap() to generate_heuristic_saliency_map() and use it only as an explicit, clearly-labeled fallback when the PyTorch checkpoint is unavailable – never silently substituted for the real one.
4. After train_dr_classifier.py finishes (once I supply the parquet data), write final validation accuracy, quadratic kappa, per-class sensitivity/specificity, and confusion matrix to TRAINING_RESULTS.md with a timestamp.
5. For detect_microaneurysms m, segment_exudates m, classify_hemorrhages m: add a second-stage classifier (shallow CNN or SVM) trained on real IDRid lesion masks to filter morphological candidates. ADDITIONALLY, for detect_microaneurysms m specifically, add sub-pixel centroid refinement: after connected-component detection, fit a 2D quadratic (paraboloid) to the intensity values in a small window around each candidate's integer-pixel centroid and solve for the true peak location to sub-pixel precision. Report both main candidate centroids (integer, existing) and main candidate centroids subpixel (float, new) in the output struct – this directly satisfies the PS's explicit "sub-pixel microaneurysm detection" requirement, which integer-pixel morphology alone does not. Write matlab/validation/evaluate_lesion_modules.m computing real Dice/IoU/sensitivity/specificity per module against the real IDRid test split, written to REAL_VALIDATION_LOG.md.
6. Write matlab/validation/evaluate_optic_disc_fovea.m: run detect_optic_disc.m and localize_fovea.m against every image in the IDRid localization subset, compute real pixel-distance error against the ground-truth OD/fovea coordinates loaded in step 2, and report mean/median error in pixels and as a percentage of image diagonal. This replaces the current "not validated" status for these two modules with a real, defensible number.
7. Write matlab/validation/evaluate_on_real_data.m: run the existing production ONNX model (inference only) against the real IDRid test split AND Messidor-2 separately, compute real sensitivity/specificity/OWK from actual confusion matrices, print both side by side to show cross-dataset generalization.
8. Write matlab/validation/ablation_study.m implementing the PS's explicit "outperforms any single technique" requirement for real:
(a) Baseline: run ONLY the EfficientNet-B3 classifier on the test set,

ignoring lesion detections and clinical arbitration — this is the "single technique "

(h) Integrated: run the full pipeline (classifier + VQID lesion detections + FTDRS 4-2-1 clinical arbitration engine) on the identical test set

Compute sensitivity, specificity, and QWK for both from real confusion matrices on the same images, and report the delta. If the integrated pipeline does not outperform the baseline, report that honestly too — it's still a real, defensible experiment either way.

9. Write matlab/benchmark_pipeline to simulink.m: run benchmark_pipeline.m, extract real per-stage timings, write them into simulink_params.m, replacing the currently assumed constants.

10. Add matlab/reporting/export_structured_report.m using the Medical Imaging Toolbox to package each screening result (grade, lesion findings, OD/fovea coordinates, confidence, Grad-CAM overlay) as a structured clinical metadata object (DICOM-SR-style key-value structure, or the closest Medical Imaging Toolbox equivalent your MATLAB version supports) alongside the existing PDF report — this directly uses the toolbox the PS lists but the repo currently ignores.

Hard rules:

- No hardcoded result strings anywhere — every printed number must come directly from a computed variable in the same scope
- No function may claim "Grad-CAM ", "validated ", "sub-pixel ", or "real" in its name, comment, or output label unless it demonstrably computes that thing — audit every existing docstring for false claims (e.g. inference nv's current "Real Score-CAM" comment) and fix or remove them
- If a step can't run because data isn't available yet, print/raise a clear "DATA MISSING — NOT RUN" state, never a placeholder result
- After each numbered task, append actual console output to REAL_VALIDATION_LOG.md with a timestamp
- Do not attempt to satisfy the PS's ">90%/>85%" language by adjusting thresholds, cherry-picking splits, or retraining on test data. Report the real result from step 7/8 as-is, even if it falls short — a short but honest number is stronger in front of judges than an unearned one.

5. Pilot Protocol for the <30-Second Claim (no code involved)

- Generate 10 full annotated reports (mix of ICDR grades 0–4) from the pipeline.
- Ask a clinically-informed reviewer (a doctor contact, mentor, or even a senior med student) to review each report cold, with a stopwatch, and decide accept/reject.
- Record each time honestly; report mean and range, explicitly labeled as a small informal pilot (N=10, one reviewer) — not a clinical validation study.
- This gives you real, defensible pilot evidence instead of an unmeasured claim, without needing a formal IRB-style trial you don't have time for.